

UXL SKILLS EVALUATOR

SYCL Build & Debug Skill

Portable build and runtime triage across UXL projects

September 2026 • 8 of 8 evaluation tasks implemented

Why this is useful across UXL projects

- Shorten recurring triage — Agents classify configure, compile, link, loader, and device-selection failures before proposing fixes.
- Capture the environment once — OS, shell, compiler, CMake, target, runtime, device, and first error travel with the report.
- Keep hypotheses separate — Compiler, package, backend, loader, device, and threading-runtime failures are tested independently.
- Route domain questions correctly — General SYCL mechanics stay shared; exact library behavior returns to its owning project.

What the skill tells an agent today

- Classify the failing phase — Configure, compile, link, runtime load, or device selection.
- Record a reproducible environment — Include toolchain versions, target, commands, first error, and a safe local probe.
- Prove the selected device — Enumerate it and run a minimal workload before blaming a library.
- Check compile-time and runtime paths separately — CMake cache, link targets, loader paths, and plugins can fail independently.
- Verify current support upstream — The skill does not embed a permanent backend matrix.

Evaluation inventory: all 8 tasks implemented

- Device and runtime — Windows/WSL discovery, loader/plugin mismatch, and silent CPU fallback.
- Build contracts — Compiler cache, backend link, compile-time backend selection, and transitive target linking.
- Runtime composition — oneDNN plus threading-runtime integration.
- What remains — Maintainer ownership and matched evidence across representative toolchains, not more task-count filling.

How we chose the content

- Project-owned sources — SYCL toolchain docs plus UXL library build, example, test, and incident evidence.
- Failure-shaped coverage — Each task starts at a boundary maintainers repeatedly diagnose.
- Portable checks — Verifiers test observed build/runtime behavior rather than vendor names.
- Honest limits — Specialized hardware enters only when hosted systems cannot reproduce the task faithfully.

What shared ownership would look like

- UXL working-group reviewers own — Shared terminology, routing rules, safe probes, and vendor-neutral limitations.
- Project maintainers validate hand-offs — Each library confirms where general SYCL triage ends and project behavior begins.
- Toolchain owners refresh sources — Compiler, loader, plugin, and package changes update the source ledger and affected tasks.
- UXL infrastructure maintains — Validators, Harbor execution, dashboards, and specialized-lane contracts.

A focused 30-minute maintainer review

- 1. Confirm the five-phase triage model — Configure, compile, link, runtime load, device selection.
- 2. Correct one routing boundary — Identify advice that belongs in a library-specific skill instead.
- 3. Judge the eight tasks — Mark smoke coverage versus tasks worth matched model trials.
- 4. Name a shared reviewer — One person or small working-group team is enough.
- Outcome — A common first-response playbook that produces reproducible reports across UXL projects.